

Exercise 4 Solutions

1.

Number of pages the process is divided

= Process size / Page size

= 256 MB / B bytes

= $2^{28} / B$

Page table size

= Number of entries in the page table x Page table entry size

= Number of pages the process is divided x Page table entry size

= $(2^{28} / B) \times 4$ bytes

= $(2^{30} / B)$ bytes

Now,

According to the above condition, we must have-

$(2^{30} / B)$ bytes $\leq$ B bytes

$B^2 \geq 2^{30}$

$B \geq 2^{15}$

Thus, minimum page size possible = 2^{15} bytes or 32 KB.

2.

In First Fit Algorithm,

	P1				
200KB	400KB	600KB	500KB	300KB	250KB

	P1	P2			
200KB	400KB	600KB	500KB	300KB	250KB

	P1	P2	P3		
200KB	400KB	600KB	500KB	300KB	250KB

P4 can not be allocated the memory.

In Best Fit Algorithm,

	P1				
200KB	400KB	600KB	500KB	300KB	250KB

	P1				P2
200KB	400KB	600KB	500KB	300KB	250KB

	P1		P3		P2
200KB	400KB	600KB	500KB	300KB	250KB

	P1	P4	P3		P2
200KB	400KB	600KB	500KB	300KB	250KB

In Worst Fit Algorithm,

		P1			
200KB	400KB	600KB	500KB	300KB	250KB

		P1	P2		
200KB	400KB	600KB	500KB	300KB	250KB

P3 and P4 can not be allocated the memory.

3.

1) FIFO page replacement algorithm

Ref String	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
	1	1	1	1	1	1	5	5	5	5	5	3	3	3	3	3	1	1	1	1
		2	2	2	2	2	2	6	6	6	6	6	7	7	7	7	7	7	3	3
			3	3	3	3	3	3	2	2	2	2	2	6	6	6	6	6	6	6
				4	4	4	4	4	4	1	1	1	1	1	1	2	2	2	2	2
Hit/Miss	M	M	M	M	H	H	M	M	M	M	H	M	M	M	H	M	M	H	M	H

Page Faults : 14 Hit Ratio: 30%

2) LRU page replacement algorithm.

Ref String	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	6	6	6	6
		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
			3	3	3	3	5	5	5	5	5	3	3	3	3	3	3	3	3	3
				4	4	4	4	6	6	6	6	6	7	7	7	7	1	1	1	1
Hit/Miss	M	M	M	M	H	H	M	M	H	H	H	M	M	M	H	H	M	H	H	H

Page Faults : 10 Hit Ratio: 50%

3) Second-chance page replacement algorithm.

Ref String	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
	1	1	1	1	1	1	5	5	5	5	5	3	3	3	3	3	1	1	1	1
		2	2	2	2	2	2	6	6	6	6	6	7	7	7	7	7	7	3	3
			3	3	3	3	3	3	2	2	2	2	2	6	6	6	6	6	6	6
				4	4	4	4	4	4	1	1	1	1	1	1	2	2	2	2	2
Hit/Miss	M	M	M	M	H	H	M	M	M	M	H	M	M	M	H	M	M	H	M	H

Page Faults : 14 Hit Ratio: 30%

4.

1) Since each page table entry is 4 bytes and each page contains 8KB, then a one-page page table would point to 2048 or 2^{11} pages, addressing a total of $2^{11} * 2^{13} = 2^{24}$ bytes.

Continuing this process:

Depth	Address Space
1	2^{24} bytes
2	2^{35} bytes
3	2^{46} bytes

Therefore, 3 levels are needed.

2) Without extra hardware, performing a memory operation takes 4 actual memory operations: 3 page table lookups in addition to the actual memory operation.

3) Multiple answers are acceptable for this question, as long as the explanations are reasonable.

1) Six physical pages totaling 48KB are needed: one for the first-level page table, one for one page of the second-level page table, one for one page of the third level page table, and three for the process' three pages.

2) At least 6 physical pages (48KB, the same as the first answer) and at most 10 physical pages are needed (80KB, assume that three pages locate far away from each other, then we need one for the first-level page table, three for three pages of the second-level page table, three for three pages of the third level page table, and three for the process' three pages), depending on the locations of data page, code page and stack page.

3) Eight physical pages totaling 64KB are needed: assume data page and code page are located close to each other, then we need one for the first-level page table, two for two pages of the second-level page table, two for two pages of the third level page table, and three for the process' three pages.

